

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      /* STAT 382 SAS HOMEWORK 1 */
70
71
72
73      /* Task 1: Data Frame Creation */
74      TITLE1 "Task 1";
75      TITLE2 "Q1 Part a) Import Datasets";
76      PROC IMPORT datafile="/home/u64394289/SAS_cars_domestic.csv"
77      OUT=domestic
78      DBMS=CSV
79      replace;
80      run;

```

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to WORK.PARMS.PARMS.SLIST.

```

81      /*****
82      *   PRODUCT:   SAS
83      *   VERSION:   9.4
84      *   CREATOR:   External File Interface
85      *   DATE:      25NOV25
86      *   DESC:      Generated SAS Datasets Code
87      *   TEMPLATE SOURCE: (None Specified.)
88      *****/
89      data WORK.DOMESTIC ;
90      %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
91      infile '/home/u64394289/SAS_cars_domestic.csv' delimiter = ',' MISSOVER DSD lrecl=32767 firstobs=2 ;
92      informat Name $18. ;
93      informat Car_Type $9. ;
94      informat Drive_Wheels_Available $12. ;
95      informat Seating best32. ;
96      informat Engine_liter best32. ;
97      informat HP best32. ;
98      informat Length_in best32. ;
99      informat Width_in best32. ;
100     informat Height_in best32. ;
101     informat Wheelbase_in best32. ;
102     informat Weight_lbs best32. ;
103     informat Cargo_Volume_cuft best32. ;
104     informat Overall_MPG best32. ;
105     informat Lowest_MSRP best32. ;
106     informat Parent_Company_HQ $3. ;
107     informat Manufacturer_Type $8. ;
108     format Name $18. ;
109     format Car_Type $9. ;
110     format Drive_Wheels_Available $12. ;
111     format Seating best12. ;
112     format Engine_liter best12. ;
113     format HP best12. ;
114     format Length_in best12. ;
115     format Width_in best12. ;
116     format Height_in best12. ;
117     format Wheelbase_in best12. ;
118     format Weight_lbs best12. ;
119     format Cargo_Volume_cuft best12. ;
120     format Overall_MPG best12. ;
121     format Lowest_MSRP best12. ;
122     format Parent_Company_HQ $3. ;
123     format Manufacturer_Type $8. ;
124     input
125         Name $
126         Car_Type $
127         Drive_Wheels_Available $
128         Seating
129         Engine_liter
130         HP
131         Length_in
132         Width_in
133         Height_in
134         Wheelbase_in
135         Weight_lbs
136         Cargo_Volume_cuft
137         Overall_MPG
138         Lowest_MSRP
139         Parent_Company_HQ $
140         Manufacturer_Type $
141     ;

```

```

142         if _ERROR_ then call symputx('_EFIERR_',1); /* set ERROR detection macro variable */
143         run;

```

NOTE: The infile '/home/u64394289/SAS_cars_domestic.csv' is:
 Filename=/home/u64394289/SAS_cars_domestic.csv,
 Owner Name=u64394289,Group Name=oda,
 Access Permission=-rw-r--r--,
 Last Modified=24Nov2025:20:46:47,
 File Size (bytes)=5086

NOTE: 57 records were read from the infile '/home/u64394289/SAS_cars_domestic.csv'.
 The minimum record length was 73.
 The maximum record length was 92.

NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	9505.43k
OS Memory	32796.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	192 Switch Count 2
Page Faults	0
Page Reclaims	172
Page Swaps	0
Voluntary Context Switches	14
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

57 rows created in WORK.DOMESTIC from /home/u64394289/SAS_cars_domestic.csv.

NOTE: WORK.DOMESTIC data set was successfully created.

NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.

NOTE: PROCEDURE IMPORT used (Total process time):

real time	0.05 seconds
user cpu time	0.03 seconds
system cpu time	0.00 seconds
memory	9505.43k
OS Memory	33056.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	192 Switch Count 10
Page Faults	0
Page Reclaims	1342
Page Swaps	0
Voluntary Context Switches	87
Involuntary Context Switches	2
Block Input Operations	0
Block Output Operations	328

```

144
145     PROC IMPORT datafile="/home/u64394289/SAS_cars_foreign.csv"
146     OUT=foreign
147     DBMS=CSV
148     replace;
149     run;

```

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to WORK.PARMS.PARMS.SLIST.

```

150     /*****
151     *   PRODUCT:   SAS
152     *   VERSION:   9.4
153     *   CREATOR:   External File Interface
154     *   DATE:      25NOV25
155     *   DESC:      Generated SAS Datastep Code
156     *   TEMPLATE SOURCE: (None Specified.)
157     *****/
158     data WORK.FOREIGN ;
159     %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
160     infile '/home/u64394289/SAS_cars_foreign.csv' delimiter = ',' MISSOVER DSD lrecl=32767 firstobs=2 ;
161     informat Name $18. ;
162     informat Car_Type $11. ;
163     informat Drive_Wheels_Available $12. ;
164     informat Seating_best32. ;
165     informat Engine_liter_best32. ;
166     informat HP_best32. ;

```

```

167      informat Length_in best32. ;
168      informat Width_in best32. ;
169      informat Height_in best32. ;
170      informat Wheelbase_in best32. ;
171      informat Weight_lbs best32. ;
172      informat Cargo_Volume_cuft best32. ;
173      informat Overall_MPG best32. ;
174      informat Lowest_MSRP best32. ;
175      informat Parent_Company_HQ $7. ;
176      informat Manufacturer_Type $7. ;
177      format Name $18. ;
178      format Car_Type $11. ;
179      format Drive_Wheels_Available $12. ;
180      format Seating best12. ;
181      format Engine_liter best12. ;
182      format HP best12. ;
183      format Length_in best12. ;
184      format Width_in best12. ;
185      format Height_in best12. ;
186      format Wheelbase_in best12. ;
187      format Weight_lbs best12. ;
188      format Cargo_Volume_cuft best12. ;
189      format Overall_MPG best12. ;
190      format Lowest_MSRP best12. ;
191      format Parent_Company_HQ $7. ;
192      format Manufacturer_Type $7. ;
193  input
194      Name $
195      Car_Type $
196      Drive_Wheels_Available $
197      Seating
198      Engine_liter
199      HP
200      Length_in
201      Width_in
202      Height_in
203      Wheelbase_in
204      Weight_lbs
205      Cargo_Volume_cuft
206      Overall_MPG
207      Lowest_MSRP
208      Parent_Company_HQ $
209      Manufacturer_Type $
210  ;
211  if _ERROR_ then call symputx('_EFIERR_',1); /* set ERROR detection macro variable */
212  run;

```

NOTE: The infile '/home/u64394289/SAS_cars_foreign.csv' is:
 Filename=/home/u64394289/SAS_cars_foreign.csv,
 Owner Name=u64394289,Group Name=oda,
 Access Permission=-rw-r--r--,
 Last Modified=24Nov2025:20:46:47,
 File Size (bytes)=14083

NOTE: 165 records were read from the infile '/home/u64394289/SAS_cars_foreign.csv'.
 The minimum record length was 69.
 The maximum record length was 98.

NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	9503.06k
OS Memory	32796.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	193 Switch Count 2
Page Faults	0
Page Reclaims	159
Page Swaps	0
Voluntary Context Switches	14
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

165 rows created in WORK.FOREIGN from /home/u64394289/SAS_cars_foreign.csv.

NOTE: WORK.FOREIGN data set was successfully created.

NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.

NOTE: PROCEDURE IMPORT used (Total process time):

real time	0.03 seconds
user cpu time	0.03 seconds
system cpu time	0.01 seconds
memory	9503.06k
OS Memory	33056.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	193 Switch Count 10
Page Faults	0
Page Reclaims	1137
Page Swaps	0
Voluntary Context Switches	82
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	288

```

213
214     TITLE2 "Q1 Part b) Modify column formats";
215     DATA domestic; *change the name to whatever you named your DOMESTIC dataset;
216         LENGTH Manufacturer_Type $30 Parent_Company_HQ $30 Car_Type $14;
217         FORMAT Manufacturer_Type $30. Parent_Company_HQ $30. Car_Type $14.;
218         SET domestic; *change the name to whatever you named your DOMESTIC dataset;
219     RUN;

```

NOTE: There were 57 observations read from the data set WORK.DOMESTIC.

NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1022.50k
OS Memory	27820.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	194 Switch Count 2
Page Faults	0
Page Reclaims	119
Page Swaps	0
Voluntary Context Switches	10
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

220
221     DATA foreign; *change the name to whatever you named your FOREIGN dataset;
222         LENGTH Manufacturer_Type $30 Parent_Company_HQ $30 Car_Type $14;
223         FORMAT Manufacturer_Type $30. Parent_Company_HQ $30. Car_Type $14.;
224         SET foreign; *change the name to whatever you named your FOREIGN dataset;
225     RUN;

```

NOTE: There were 165 observations read from the data set WORK.FOREIGN.

NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	959.62k
OS Memory	27820.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	195 Switch Count 2
Page Faults	0
Page Reclaims	114
Page Swaps	0
Voluntary Context Switches	10
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	272

```

226
227
228
229     TITLE2 "Q1 Part c) Combine DOMESTIC and FOREIGN Datasets"; * list DOMESTIC first;
230     data all_cars;
231         set domestic foreign;
232     run;

```

NOTE: There were 57 observations read from the data set WORK.DOMESTIC.

```
NOTE: There were 165 observations read from the data set WORK.FOREIGN.
NOTE: The data set WORK.ALL_CARS has 222 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1357.40k
      OS Memory           28080.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count          196   Switch Count   2
      Page Faults         0
      Page Reclaims       145
      Page Swaps          0
      Voluntary Context Switches 11
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264
```

```
233
234
235      TITLE2 "Q1 Part d) Sort all_cars by ascending Drive_Wheels_Available";
236      proc sort data = all_cars;
237      by Drive_Wheels_Available;
238      run;
```

```
NOTE: There were 222 observations read from the data set WORK.ALL_CARS.
NOTE: The data set WORK.ALL_CARS has 222 observations and 16 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              926.31k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count          197   Switch Count   2
      Page Faults         0
      Page Reclaims       103
      Page Swaps          0
      Voluntary Context Switches 13
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264
```

```
239
240
241      TITLE2 "Q1 Part e) Create the dataset DriveWheels";
242      data DriveWheels;
243      length Drive_Wheels_Available $12 Drive_Wheels_Max $10;
244      input Drive_Wheels_Available $1-12 Drive_Wheels_Max $14-23;
245      datalines;
```

```
NOTE: The data set WORK.DRIVEWHEELS has 9 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              739.84k
      OS Memory           27560.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count          198   Switch Count   2
      Page Faults         0
      Page Reclaims       87
      Page Swaps          0
      Voluntary Context Switches 12
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264
```

```
255      ;
256      run;
257
258      proc print data = DriveWheels;
259      run;
```

```
NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.01 seconds
```

```

user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory             1375.96k
OS Memory          27560.00k
Timestamp          11/25/2025 03:58:24 AM
Step Count         199   Switch Count  0
Page Faults        0
Page Reclaims      62
Page Swaps         0
Voluntary Context Switches  0
Involuntary Context Switches 1
Block Input Operations  0
Block Output Operations  8

```

```

260
261     proc contents data = DriveWheels;
262     run;

```

NOTE: PROCEDURE CONTENTS used (Total process time):

```

real time          0.02 seconds
user cpu time      0.02 seconds
system cpu time    0.01 seconds
memory             1334.37k
OS Memory          27820.00k
Timestamp          11/25/2025 03:58:24 AM
Step Count         200   Switch Count  0
Page Faults        0
Page Reclaims      130
Page Swaps         0
Voluntary Context Switches  3
Involuntary Context Switches 2
Block Input Operations  0
Block Output Operations  8

```

```

263
264     TITLE2 "Q1 Part f) Sort DriveWheels by ascending Drive_Wheels_Available";
265     proc sort data = DriveWheels;
266     by Drive_Wheels_Available;
267     run;

```

NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.

NOTE: The data set WORK.DRIVEWHEELS has 9 observations and 2 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory             932.87k
OS Memory          27820.00k
Timestamp          11/25/2025 03:58:24 AM
Step Count         201   Switch Count  2
Page Faults        0
Page Reclaims      114
Page Swaps         0
Voluntary Context Switches  12
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  264

```

```

268
269
270
271     TITLE2 "Q1 Part g) Merge the all_cars dataset and the DriveWheels dataset to create all_cars2";
272     data all_cars2;
273     merge all_cars DriveWheels;
274     by Drive_Wheels_Available;
275     run;

```

NOTE: There were 222 observations read from the data set WORK.ALL_CARS.

NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.

NOTE: The data set WORK.ALL_CARS2 has 222 observations and 17 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory             1405.62k
OS Memory          28080.00k
Timestamp          11/25/2025 03:58:24 AM

```

Step Count	202	Switch Count	2
Page Faults	0		
Page Reclaims	144		
Page Swaps	0		
Voluntary Context Switches	14		
Involuntary Context Switches	0		
Block Input Operations	0		
Block Output Operations	264		

276
277
278
279
280
281
282
283
284
285
286
287
288
289
290

```

/* Task 2: Cleaning and Formatting */
TITLE1 "Task 2";
TITLE2 "Q2) Replace missing values for Engine_liter with 0s in new dataset cars_fix";
data cars_fix;
set all_cars2;
if Engine_liter = . then Engine_liter = 0;
run;

```

NOTE: There were 222 observations read from the data set WORK.ALL_CARS2.
NOTE: The data set WORK.CARS_FIX has 222 observations and 17 variables.
NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	959.12k
OS Memory	27820.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	203
Page Faults	0
Page Reclaims	110
Page Swaps	0
Voluntary Context Switches	13
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

291
292
293
294
295
296
297
298
299

```

TITLE2 "Q3) Remove rows for Car_Type and Overall_MPG that are missing. Do this in a new dataset called cars_clean";
data cars_clean;
set cars_fix;
if Car_Type = "" then delete;
if Overall_MPG = . then delete;
run;

```

NOTE: There were 222 observations read from the data set WORK.CARS_FIX.
NOTE: The data set WORK.CARS_CLEAN has 216 observations and 17 variables.
NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1020.53k
OS Memory	28076.00k
Timestamp	11/25/2025 03:58:24 AM
Step Count	204
Page Faults	0
Page Reclaims	113
Page Swaps	0
Voluntary Context Switches	10
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	264

300
301
302
303
304
305

```
306
307
308     OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
318
```